

Probability

COMPLETE LEARNING GUIDE

What You'll Learn

Probability questions in placement tests build directly on permutations and combinations, framing 'favorable outcomes ÷ total outcomes' around familiar objects — coins, dice, cards, and colored balls in a bag — while testing whether a candidate can correctly distinguish independent from dependent events and mutually exclusive from overlapping events. The most frequent point of failure is applying the same probability to a second draw when items are NOT replaced (dependent events), or forgetting to subtract the overlap when two events can happen simultaneously. This sub-module covers the classical probability definition and sample space counting, dice and card problems, the addition rule for overlapping events, conditional probability with and without replacement, solving probability problems via combinations, and the complementary-counting shortcut for 'at least one' probability questions — each with a fully worked numerical example.

1. Classical Probability and Sample Space

Probability of an event = (number of favorable outcomes) ÷ (total number of equally likely outcomes). The first step in any probability problem is correctly identifying the full sample space before counting favorable cases within it.

- A fair die is rolled once. Find the probability of getting a number greater than 4. Favorable outcomes = {5, 6}, total outcomes = {1, 2, 3, 4, 5, 6}. $P = 2/6 = 1/3$.

2. Probability With a Standard Deck of Cards

A standard deck has 52 cards: 4 suits (hearts, diamonds, clubs, spades) of 13 cards each, including 1 ace, 3 face cards (jack, queen, king), and 9 number cards per suit. Memorizing this structure makes card-probability questions fast to set up.

- A card is drawn at random from a well-shuffled deck. $P(\text{drawing an ace}) = 4/52 = 1/13$.
 $P(\text{drawing a face card}) = 12/52 = 3/13$ (3 face cards × 4 suits).

3. Probability With Dice

When two dice are rolled, the sample space has 36 EQUALLY LIKELY ordered outcomes (6×6), and each specific sum must be counted by listing every ordered pair that produces it — (2, 6) and (6, 2) are counted as two separate outcomes, not merged into one.

- Two fair dice are rolled. Find $P(\text{sum} = 8)$. Favorable ordered pairs: (2,6),(3,5),(4,4),(5,3), (6,2) — 5 outcomes. $P = 5/36$.

4. The Addition Rule for Overlapping Events

For two events A and B that can occur together (are not mutually exclusive), $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$ — the overlap must be subtracted once to avoid double-counting. If A and B cannot occur together (mutually exclusive), the overlap term is simply zero.

- A card is drawn from a deck. Find $P(\text{king or heart})$. $P(\text{king}) = 4/52$, $P(\text{heart}) = 13/52$, $P(\text{king of hearts, the overlap}) = 1/52$. $P(\text{king or heart}) = 4/52 + 13/52 - 1/52 = 16/52 = 4/13$.

5. Conditional Probability: With and Without Replacement

When items are drawn WITHOUT replacement, each subsequent draw's probability changes because the pool of remaining items has changed — multiply the sequence of changing probabilities. When items are drawn WITH replacement, the pool resets each time, so the same probability applies to every draw.

- A bag has 5 red and 3 blue balls (8 total). Two balls are drawn WITHOUT replacement. $P(\text{both red}) = (5/8) \times (4/7) = 20/56 = 5/14$.

- Same bag, but drawn WITH replacement. $P(\text{both red}) = (5/8) \times (5/8) = 25/64$ — noticeably different from the without-replacement case.

6. Solving Probability Using Combinations

For 'both/all specific items' probability questions, it's often faster to skip the sequential multiplication and instead compute $P = C(\text{favorable}, r) \div C(\text{total}, r)$ directly — selecting r items from the favorable group divided by selecting r items from everything, which gives the same answer as the without-replacement sequential method.

- Same bag (5 red, 3 blue). $P(\text{both drawn balls are red})$ via combinations $= C(5,2)/C(8,2) = 10/28 = 5/14$ — matches the sequential without-replacement calculation exactly, confirming both methods are equivalent.

7. 'At Least One' Probability via the Complement

For 'at least one success' probability questions, compute $1 - P(\text{no successes at all})$ rather than summing $P(\text{exactly } 1) + P(\text{exactly } 2) + \dots$ — the complement approach is almost always faster and less error-prone.

- A fair coin is tossed 3 times. Find $P(\text{at least one head})$. $P(\text{no heads at all, i.e., all 3 tails}) = (1/2)^3 = 1/8$. $P(\text{at least one head}) = 1 - 1/8 = 7/8$.

Common Pitfalls

- Forgetting to subtract the intersection $P(A \text{ and } B)$ when events overlap, leading to double-counting in $P(A \text{ or } B)$.
- Applying the same probability to every draw (as in 'with replacement') when the question specifies items are drawn without replacement, where probabilities must change each draw.
- Merging symmetric ordered dice outcomes like (2,6) and (6,2) into a single outcome, undercounting the true sample space of 36.
- Confusing independent events (multiply raw probabilities directly) with dependent events (the second event's probability depends on the first event's outcome).
- Mixing sequential multiplication and combination-based methods within the same calculation instead of using one consistent approach.
- Not using the complement ($1 - P(\text{none})$) for 'at least one' questions, instead attempting a much longer direct case-by-case enumeration.
- Forgetting the base counts: a standard deck has 52 cards, a single die has 6 faces, and two dice together have 36 ordered outcomes.

Key Takeaways

- $P(\text{event}) = \text{favorable outcomes} \div \text{total equally likely outcomes in the full sample space}$.
- $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$ whenever the events can occur together.
- Without replacement: multiply probabilities that change each draw; with replacement: the same probability applies to every draw.
- $P(\text{specific group of items}) \text{ via combinations} = C(\text{favorable}, r) \div C(\text{total}, r)$ — equivalent to, and often faster than, sequential multiplication.
- 'At least one' probability = $1 - P(\text{none})$ — always faster than summing every individual success case.
- Memorize base sample space sizes: 52-card deck, 6-face die, 36 ordered outcomes for two dice.

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GeeksforGeeks — Probability Aptitude Questions

<https://www.geeksforgeeks.org/maths/probability-questions/>

IndiaBix — Probability Aptitude Questions

<https://www.indiabix.com/aptitude/probability/>